

```
In [1]: import boto3
import pandas as pd
from sklearn.cluster import KMeans
```

```
In [2]: s3 = boto3.client('s3')
bucket = 'cust-seg-ygp'
key = 'rfm_scaled.csv' # <-- scaled file name

obj = s3.get_object(Bucket=bucket, Key=key)
rfm_scaled = pd.read_csv(obj['Body'])

rfm_scaled.head()
```

```
Out[2]:
```

	Customer ID	Recency_scaled	Frequency_scaled	Monetary_scaled
0	12346	0.595584	0.438998	5.166378
1	12347	-0.952279	0.131502	0.136127
2	12348	-0.603532	-0.099120	-0.064857
3	12349	-0.871064	-0.175994	0.101996
4	12350	0.519146	-0.406616	-0.181549

```
In [3]: # X = your scaled RFM features
X = rfm_scaled[['Recency_scaled', 'Frequency_scaled', 'Monetary_scaled']]

# create K-Means model with K = 4
kmeans = KMeans(n_clusters=4, random_state=42)

# fit the model and get cluster labels
rfm_scaled['Cluster'] = kmeans.fit_predict(X)

# preview results
rfm_scaled.head()
```

```
Out[3]:
```

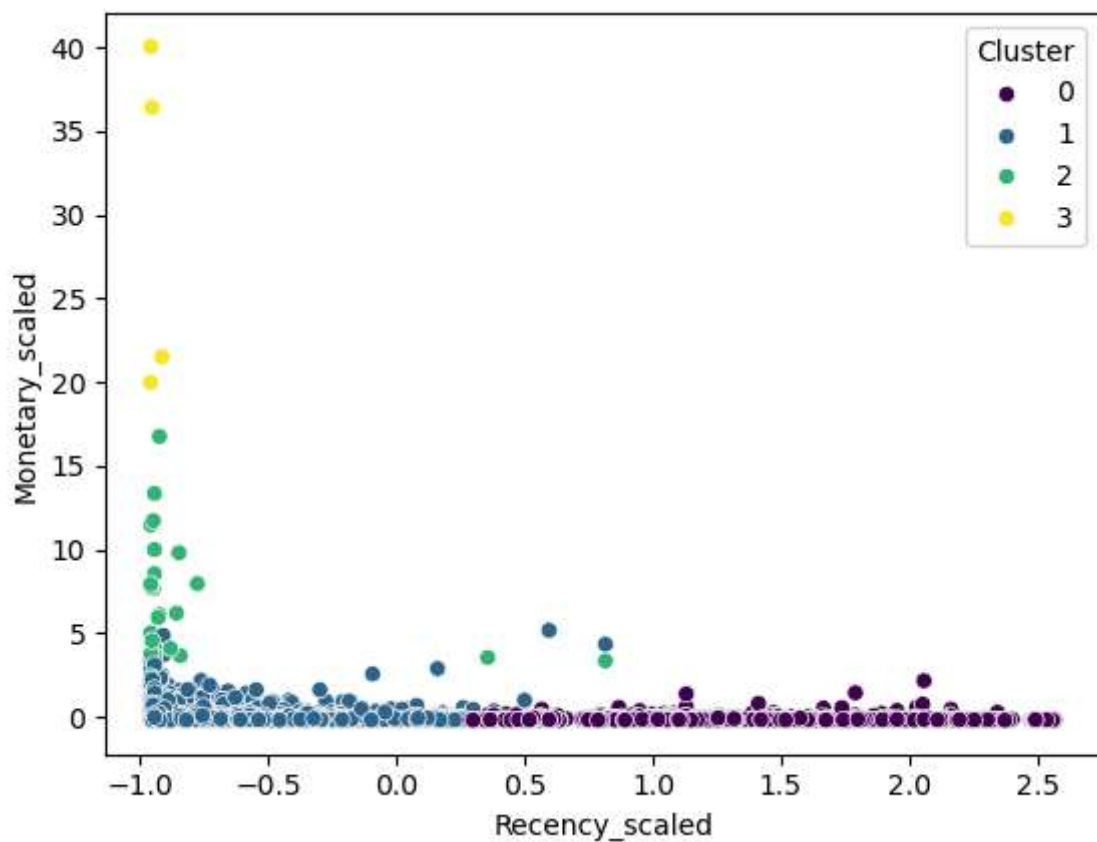
	Customer ID	Recency_scaled	Frequency_scaled	Monetary_scaled	Cluster
0	12346	0.595584	0.438998	5.166378	1
1	12347	-0.952279	0.131502	0.136127	1
2	12348	-0.603532	-0.099120	-0.064857	1
3	12349	-0.871064	-0.175994	0.101996	1
4	12350	0.519146	-0.406616	-0.181549	0

```
In [5]: rfm_scaled['Cluster'].value_counts()
```

```
Out[5]: Cluster
1      3836
0      2003
2        35
3         4
Name: count, dtype: int64
```

```
In [6]: import seaborn as sns
import matplotlib.pyplot as plt

sns.scatterplot(
    data=rfm_scaled,
    x='Recency_scaled',
    y='Monetary_scaled',
    hue='Cluster',
    palette='viridis'
)
plt.show()
```



```
In [8]: # Mapping old cluster numbers to new names
cluster_map = {
    1: 'A_Regular_Customers',
    0: 'B_Inactive_Customers',
    2: 'C_Good_Customers',
    3: 'D_VIP_Customers'
}

# Create a new column with renamed clusters
rfm_scaled['Cluster_Name'] = rfm_scaled['Cluster'].map(cluster_map)
```

```
rfm_scaled[['Cluster', 'Cluster_Name']].head()
```

Out[8]:

	Cluster	Cluster_Name
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0	1	A_Regular_Customers
1	1	A_Regular_Customers
2	1	A_Regular_Customers
3	1	A_Regular_Customers
4	0	B_Inactive_Customers

```
In [7]: import io
csv_buffer = io.StringIO()

rfm_scaled.to_csv(csv_buffer, index=False)

s3.put_object(
    Bucket='cust-seg-ygp',
    Key='rfm_scaled_clusters_k4.csv',
    Body=csv_buffer.getvalue()
)

print("Saved rfm_scaled_clusters_k4.csv to S3.")

Saved rfm_scaled_clusters_k4.csv to S3.
```

In []: